Where we are, and what is pending your decision

A short map so the open questions are clean while you think through the 4b corner cases.

Settled / in place

- **4a** (equivocation prevention) is live on all 100 committee nodes: a producer holding a quorum-certified block at a height whose anchor is being restored after a reorg-of-reorg neither proposes nor backs an alternative there; it waits for the previous best chain to return.
- **VerifySeqLegSafe now also requires quorum certification**, not just anchoring (a swap leg's block must be immediately final AND anchored at/after the counter-leg). No minimum-anchor-depth wait; the anchor axis stays 0-conf, so real-time anchoring is intact.
- **The VRF is reliable** for sortition, leader selection, and voting, and the leader VRF score is already the implemented same-height fork-choice tiebreak (my earlier "unreliable" was a misread of a comment about VRF-as-a-sortition-seed, not fork choice).

None of the above depends on the pending item below.

Pending your decision: the 4b reconciliation rule

The one open consensus question is whether (and how) to add an automatic convergence rule for the RESIDUAL case, and its exact shape. To frame your corner-case thinking, here is the decision laid out.

The regime it targets. 4a plus 4b's strictly-above rule already heal every ASYMMETRIC split (the majority keeps producing quorum blocks above the fork; the minority yields; the minority can never make another quorum). The residual is the SYMMETRIC case: roughly half the committee on each side, neither ever reaching quorum, both sides advancing sub-threshold via escaping-stall. That is governed by the fork-choice comparator, not by 4b and not by the finality gate (sub-quorum blocks are never immediately final).

Corner cases worth pressure-testing before you commit:

1. **Is a rule even needed, or is the comparator enough?** In the symmetric split the comparator ties on signature power; the next key, the leader VRF score, breaks it deterministically the moment nodes see both chains, EXCEPT in two pathological cases: the VRF score is unset on both (VRF disabled, or a block with no valid VRF proof) or the same leader signed both (equivocation). Only there does the code drop to node-local first-seen, which cannot converge a partition. So the minimal fix may be just: replace that last first-seen step with a content-derived key (lowest block hash). Question for you: is lowest-hash acceptable as the ultimate backstop, given that the only party who can grind it is one that already controls ~half the

committee (i.e. already in the compromised regime)?

2. **The reconvergence double-spend.** Any rule that lets a node abandon a chain can, in principle, reorg a swap that settled on the losing side. The efficient version of that attack is a majority-STAKE attack with ZERO Bitcoin hashpower (root cause B: both chains re-anchor to the same valid Bitcoin block, so anchoring cannot arbitrate and Bitcoin is left untouched so the Bitcoin leg stands). A cheaper sub-majority variant needs an induced stall. Question: what safety boundary is acceptable here, given we already assume honest-majority stake?
3. **The safety coupling is now in place.** Because VerifySeqLegSafe requires quorum certification, a swap can no longer settle on a sub-quorum (reorgable) block, which closes the reconvergence double-spend against swaps by construction, independent of whatever 4b rule you choose. Worth keeping in mind: the swap-safety hole and the convergence rule are now decoupled.
4. **A finality-overriding timer.** I would advise against any "after N Bitcoin blocks, force reconvergence" rule that can reorg an immediately-final block: it turns a rare, detectable, fail-safe stall into an attacker-inducible reorg of final history (inducible with zero Bitcoin hashpower via a partition), and it reopens the same-height comparison you deliberately withdrew. The deterministic-tiebreak fix in (1) handles the sub-quorum case without touching finality. But this is your call, and it is exactly the kind of corner case you are right to sit with.

My recommendation, for when you are ready: the smallest safe change is (1), the deterministic content-derived backstop for the last comparator step, applied only where signature power AND the VRF score both tie. Everything else (finality-overriding timers, same-height certificate comparison) I would leave out. But I will not implement anything here until you have finished with the corner cases and told me the shape you want.

Next

Separately, we are about to open the Simplicity-in-the-DEX thread (covenant-enforced offers on the Sequentia leg; the Bitcoin leg stays interactive). That is independent of the above and does not touch the 4b decision.